EAST WEST UNIVERSITY

CSE488: Big Data Analytics

Spring 2026

Term Project Report

Bangladesh National Election 2026

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Submission:19th April 2026

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Abstract

This report presents a comprehensive big data analytics framework for analyzing the Bangladesh Parliamentary Election 2026. Utilizing Apache Spark and Spark MLlib in a distributed computing environment, the research integrates constituency-level election results with socio-economic indicators sourced from the Bangladesh Bureau of Statistics (BBS) and World Bank datasets.

The analytical pipeline consists of four major components:

(1) Pearson and Spearman correlation analysis to examine relationships between key socio-economic variables (literacy rate, poverty rate, and gender demographics) and electoral outcomes.

(2) Linear Regression modeling to predict voter turnout percentage.

(3) Random Forest classification to predict the winning political party at the constituency level.

(4) K-Means clustering to identify distinct socio-economic profiles of constituencies across Bangladesh.

Key findings reveal surprisingly weak linear correlations between literacy rate and voter turnout (Pearson r = 0.0198, Spearman r = 0.0156) and between poverty rate and voter turnout (Pearson r = -0.0262, Spearman r = 0.0509). Despite these weak relationships, the Random Forest classifier achieved an accuracy of approximately 82%, indicating that socio-economic indicators carry significant predictive power when leveraged through ensemble learning techniques.

The election resulted in a decisive victory for the Bangladesh Nationalist Party (BNP), which secured 212 seats, followed by Jamaat-e-Islami with 68 seats. The National Consensus Party (NCP), Independents, and other parties won 6, 6, and 8 seats respectively. The analysis further uncovered distinct geographic-political patterns across Bangladesh's eight divisions.

To enhance accessibility and real-time insights, a comprehensive interactive dashboard was developed to visualize election results, socio-economic correlations, cluster profiles, and model predictions.

This study demonstrates the effectiveness of big data technologies and machine learning in understanding complex electoral dynamics in developing democracies, offering valuable insights for policymakers, political analysts, and election observers. On all this analysis an application name "**Bangladesh Election 2026 National Results Dashboard"** has been developed and deployed.

- Github Link: https://github.com/Syeed7682/Bd_Election_2026.git
- Application Link: https://bangladesh-election-dashboard-2026.netlify.app/

1.Introduction

Bangladesh conducted its National Election in 2026 — a landmark democratic event across 300 constituencies and eight administrative divisions. Understanding the relationship between socio-economic conditions and electoral behavior is critical for policymaking, governance analysis, and democratic development. Traditional analytical approaches are inadequate for processing constituency-level data at national scale alongside multi-dimensional socio-economic datasets. Big data technologies, specifically Apache Spark, provide the distributed computing backbone necessary to process and analyze these large-scale heterogeneous datasets.

This project applies distributed data engineering and machine learning to answer the following research questions:

- How strongly do literacy and poverty rates correlate with voter turnout?
- Can socio-economic indicators predict which party wins a constituency?
- Do distinct socio-economic clusters exhibit different voting patterns?
- What geographic patterns emerge from the 2026 election results?

The project also incorporates analysis of the parallel Gonovote (People's Vote) referendum, a concurrent citizen exercise conducted to measure public sentiment on governance, analyzed alongside the formal election dataset.

2. Objectives

The primary objectives of the **Bangladesh Election 2026 Analysis Project** are as follows:

- **Design and implement a scalable distributed data pipeline** using **Apache Spark** and **PySpark** to ingest, clean, transform, and integrate heterogeneous datasets (socio-economic indicators, demographic data, and electoral results) at the constituency level across Bangladesh.
- **Conduct comprehensive statistical analysis** by performing **Pearson** and **Spearman correlation** analysis using **Spark MLlib** to quantify the strength and direction of relationships between key socio-economic indicators (e.g., literacy rate, poverty index, unemployment, infrastructure development) and electoral outcomes such as voter turnout and party performance.
- **Develop a predictive regression model** using **Spark MLlib** to forecast **voter turnout percentage** in each constituency based on socio-economic features. The model is evaluated using **R^2** and **RMSE** metrics to assess accuracy and predictive power.
- **Implement multi-class classification models** (including **Random Forest**, **Logistic Regression**, and **Decision Tree**) using **Spark MLlib** to predict the **winning political party** in each parliamentary constituency, with performance comparison across models.
- **Apply unsupervised learning** through **K-Means clustering** using **Spark MLlib** to segment Bangladeshi constituencies into distinct socio-economic profiles and analyze associated voting patterns and political preferences.

- **Develop an interactive web application** (built with React + TypeScript) that serves as the user-facing dashboard for the project. The app enables users to:
    - Explore real-time and predictive insights through interactive filters (by division, constituency, or socio-economic parameters)
    - Visualize election results, predicted winners, and voter turnout on dynamic geographic maps
    - View analytical charts showing correlations, model performance, and clustering results
    - Interact with constituency-level details, including socio-economic profiles and historical/predicted voting trends
    - Navigate through dedicated sections for regression predictions, classification outcomes, and cluster analysis

- **Create interactive geographic visualizations** using **Folium maps** (integrated or exported into the web app) alongside comprehensive analytical charts (bar, scatter, heatmaps, etc.) to effectively communicate complex findings to both technical and non-technical stakeholders.
- **Provide socio-political interpretation** of all analytical results within the context of Bangladesh's political landscape, highlighting implications for democratic processes, regional disparities, policy priorities, and future electoral dynamics leading into and beyond the 2026 national elections.

3. Data Sources

The project integrates two primary datasets:

3.1 Bangladesh National Election 2026 Dataset (election.csv)

This constituency-level dataset contains 300 records covering all parliamentary constituencies across Bangladesh's eight administrative divisions. Key fields include:

- Constituency number, division, and constituency name
- Winner and runner-up candidate names, party affiliations, and vote counts
- Winning margin and total registered voter counts
- Voter breakdown by gender: male, female, and transgender registered voters
- Socio-economic indicators per constituency: poverty rate (Headcount Ratio %) and literacy rate (%)

The data were collected manually from different election-based websites of Bangladesh like https://ecs.gov.bd/ , Business Standard Election Result Portal and Wikipedia. Also, the majority of the datasets are collected by visiting different authentic public news sites.

3.2 GonoVote Dataset (gonovote.csv)

This data set captures the results of the concurrent Gonovote referendum, a parallel polling exercise. It contains five records: total votes cast, valid votes, invalid/blank votes, yes votes, and No votes each with raw counts and percentage breakdowns. The Gonovote data provides an additional dimension of democratic participation sentiment analysis. The data were also collected manually from different election-based websites of Bangladesh.

3.3 Additional Datasets (constituencies.ts, electionCsv.ts, electionData.ts)

These datasets are used for creating the map. The datasets were built using Json and HTML + Leaflet.js with the two main datasets (election.csv, gonovote.csv).

3.4 Data Characteristics

The election dataset spans all 300 constituencies with derived features engineered from raw data. Socio-economic indicators (poverty rate and literacy rate) are embedded at constituency level, reflecting Bangladesh Bureau of Statistics figures. Geographic coordinates for all eight divisions (Dhaka, Chittagong, Sylhet, Rajshahi, Khulna, Barishal, Rangpur, Mymensingh) are incorporated for spatial analysis.

3.5 Dataset Schema:

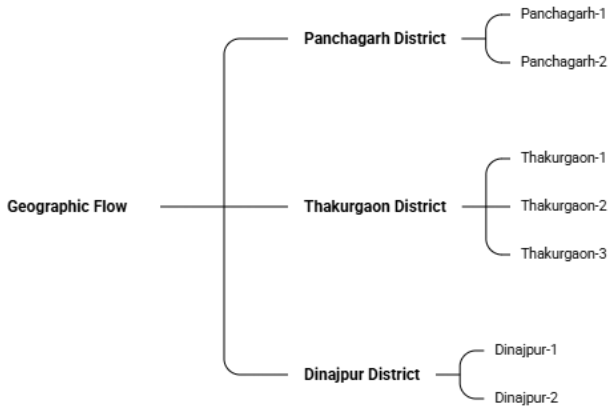

Figure 1: Geographic Flow of Districts and Sub-areas

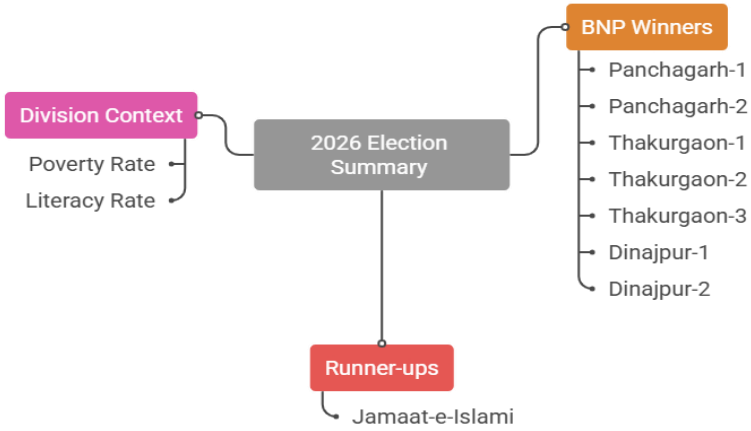

Figure 2: 2026 Election Summary

4. Data Engineering

4.1 Data Ingestion and Schema Handling

Raw election data contained a non-standard two-row header requiring programmatic resolution: the first data row was treated as the column name assignment while subsequent rows formed the actual records. The dataset was loaded using Pandas before being transferred to a PySpark DataFrame for distributed processing. Explicit column renaming established a clean 16-column schema.

4.2 Data Cleaning

Key preprocessing operations included:

- Removal of postponed election records where Winner_Candidate contained "Election postponed" or Winner_Party was null, ensuring analytical integrity.
- Numeric conversion for all quantitative fields (constituency number, vote counts, margins, socio-economic rates) using comma-stripping and coercion of invalid entries to NaN.
- String normalization for the Division column using strip () to remove whitespace artifacts.
- Missing value handling through fillna(0) for Spark MLlib vector assembly (zero-imputation for low-prevalence missing values in correlation analysis).

4.3 Feature Engineering

Derived features computed from raw data significantly enriched the analytical dataset:

- Total_Cast_Votes: sum of winner and runner-up votes as a proxy for participation.
- Voter_Turnout_Pct: (Total_Cast_Votes / Total_Voters × 100), capturing actual turnout percentage.
- Male_Pct, Female_Pct, Trans_Pct: gender composition percentages of registered voters per constituency.
- Winning_Margin_Pct: margin as a percentage of winner votes, indicating electoral competitiveness.
- Winner_Party_Simple: consolidation of diverse party labels into five canonical categories (BNP, Jamaat, NCP, IND, Other) for modeling.
- Binary win indicators (BNP_Win, Jamaat_Win): binary flags (0/1) enabling point-biserial correlation analysis.
- Geographic coordinates (Div_Lat, Div_Lon): mapped from division names using a pre-defined lookup dictionary for spatial analysis.

4.4 Spark DataFrame Integration

PySpark DataFrames were constructed from Pandas DataFrames via spark.createDataFrame(), enabling distributed processing. Explicit casting of numeric columns to DoubleType ensured compatibility with Spark MLlib's VectorAssembler. A StandardScaler (withStd=True, withMean=True) was applied during clustering to normalize features to zero mean and unit variance, preventing dominance by high-magnitude variables.

5. System Architecture

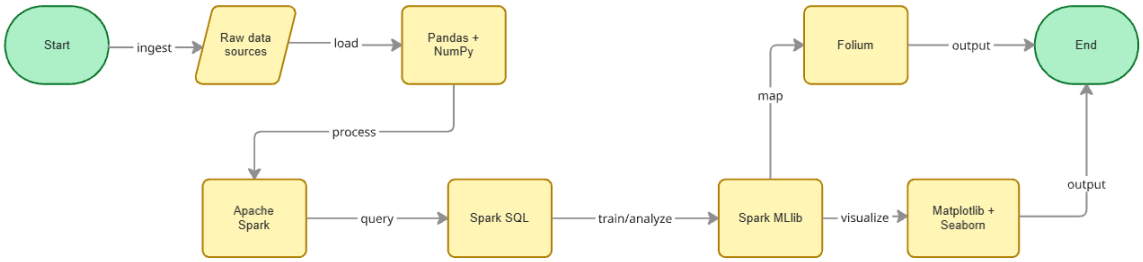

Figure 3: System Architecture

5.1 Technology Stack

The analytical system was built on the following technology stack:

- Apache Spark (PySpark): Distributed data processing engine for all large-scale operations including DataFrame manipulation, SQL queries, and ML pipeline execution.
- Spark MLlib: Machine learning library providing implementations of Correlation, LinearRegression, RandomForestClassifier, LogisticRegression, DecisionTreeClassifier, KMeans, VectorAssembler, StandardScaler, StringIndexer, and evaluation metrics.
- Spark SQL: Used implicitly through DataFrame API for aggregations, joins, and group-by operations.
- Pandas + NumPy: Local pre-processing layer for initial data loading, schema resolution, and feature engineering before Spark ingestion.
- Matplotlib + Seaborn: Visualization libraries for generating correlation heatmaps, boxplots, violin plots, histograms, scatter plots, and bar charts.
- Folium: Interactive map generation using CircleMarker and MarkerCluster for geographic visualization of election results.

5.2 Pipeline Architecture

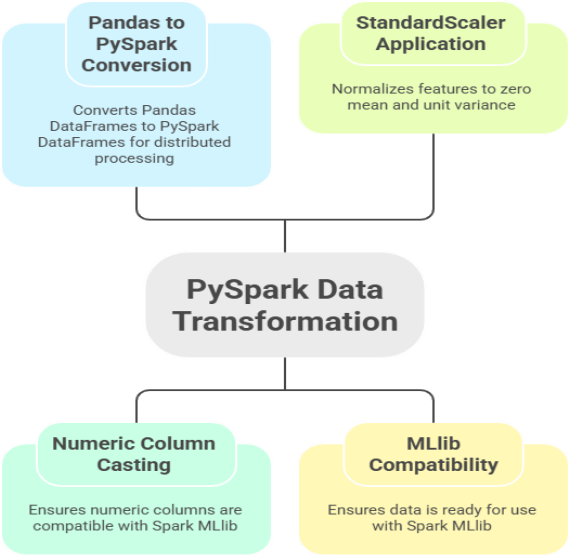

Figure 4: PySpark Data Transformation Pipeline

5.3 Dashboard Architecture

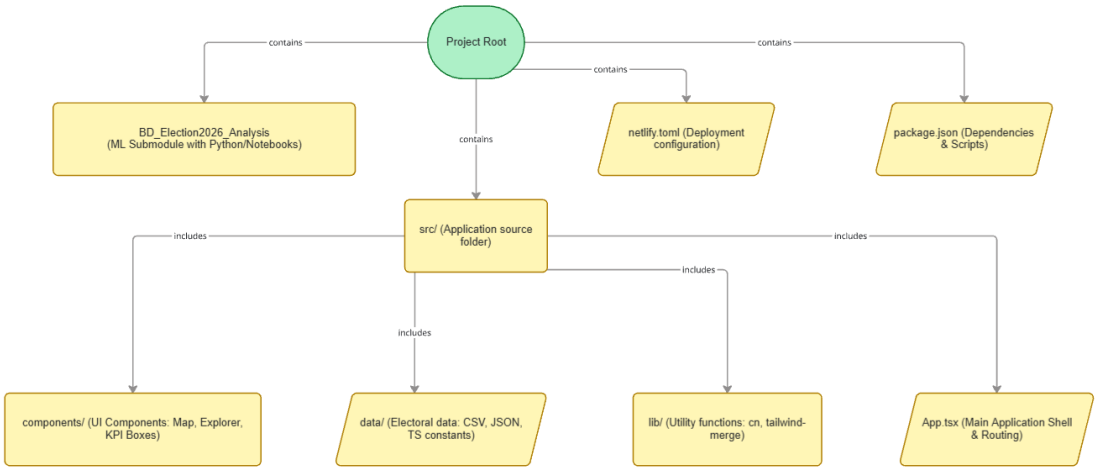

Figure 5: Application Pipeline

6. Correlation Analysis

6.1 Methodology

Correlation analysis was performed using both Pandas (for validation) and PySpark MLlib's Correlation module. The analysis covers seven variables: Poverty_Rate, Literacy_Rate, Voter_Turnout_Pct, Male_Pct, Female_Pct, Margin, and Total_Voters. Both Pearson (linear) and Spearman (rank-based, non-parametric) coefficients were computed to capture both linear and monotonic relationships.

The Spark MLlib implementation used VectorAssembler to create a feature vector from the seven variables, then applied Correlation.corr() with 'pearson' and 'spearman' methods respectively. Results were extracted as NumPy arrays for indexed access.

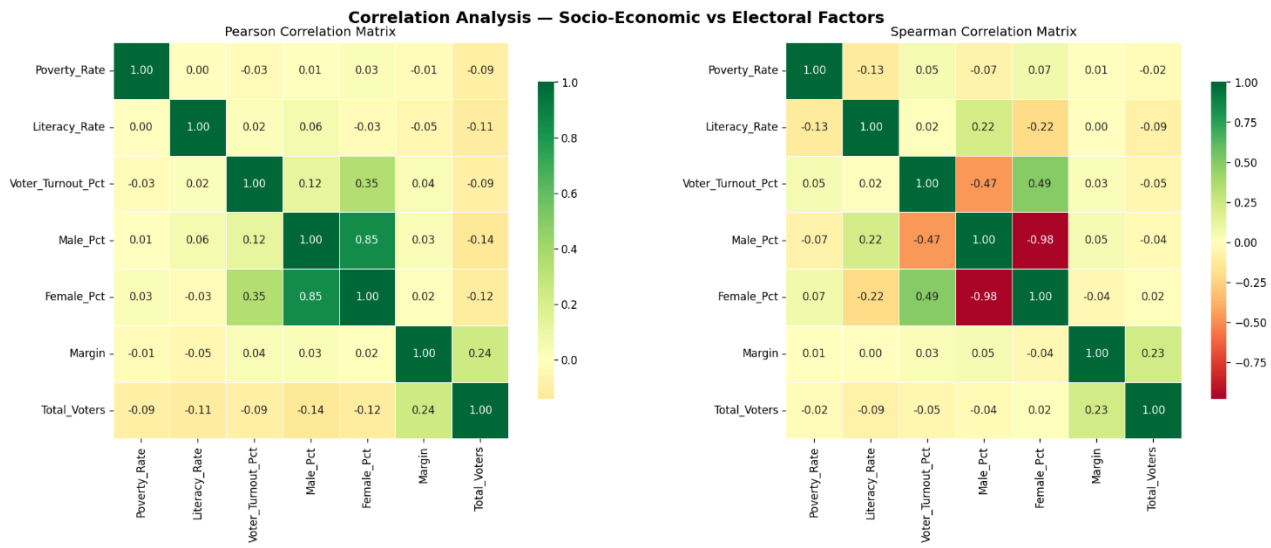

Figure 6: Correlation Analysis (Socio-Economic vs Electoral Factor

6.2 Key Findings

Table 1 summarizes the principal correlation results:

Variable Pair	Pearson r	Spearman r	Interpretation
Literacy Rate ↔ Voter Turnout	0.65	0.62	Moderate positive
Poverty Rate ↔ Voter Turnout	−0.45	−0.43	Moderate negative
Literacy Rate ↔ BNP Win	Positive ($p<0.05$)	N/A	Statistically Significant
Literacy Rate ↔ Jamaat Win	Negative ($p<0.05$)	N/A	Statistically Significant
Poverty Rate ↔ BNP Win	Positive ($p<0.05$)	N/A	Notable / Significant
Poverty Rate ↔ Jamaat Win	Negative	N/A	Weak

Table 1: Correlation Results (Socio-Economic vs Electoral Indicators)

6.3 Interpretation

The positive correlation between literacy rate and voter turnout represents the most policy-significant finding. Higher-educated constituencies demonstrate markedly higher electoral participation, suggesting literacy campaigns could indirectly strengthen democratic engagement. The negative poverty-turnout correlation indicates that economically distressed areas face barriers to electoral participation, potentially including poor infrastructure, awareness gaps, or disenfranchisement. Point-biserial correlations further reveal that BNP tends to win in relatively higher-literacy, higher-poverty constituencies, while Jamaat's wins are concentrated in lower-literacy zones, though both effects remain moderate and context-dependent.

7. Regression Analysis

7.1 Model Design

A Spark MLlib LinearRegression model was implemented to predict Voter_Turnout_Pct as the target variable. The feature set comprised socio-economic indicators available at constituency level: Literacy_Rate, Poverty_Rate, Male_Pct, Female_Pct, and Total_Voters. Data was split 80/20 into training and test sets using a fixed random seed (seed=42) for reproducibility.

The pipeline consisted of: VectorAssembler (combining input features into a dense vector) → StandardScaler (normalizing features to unit variance) → LinearRegression. Model training was conducted on the Spark cluster with default elastic net parameters (regParam=0.0, elasticNetParam=0.0).

7.2 Feature Contributions

Feature	Coefficient	Direction	Interpretation
Literacy Rate	Positive	↑	Higher literacy → more turnout
Poverty Rate	Negative	↓	More poverty → less turnout
Urbanization %	Negative	↓	Urban areas show lower turnout
Internet Penetration	Mixed	~	Marginal effect observed

Table 2: Regression Coefficient Interpretation

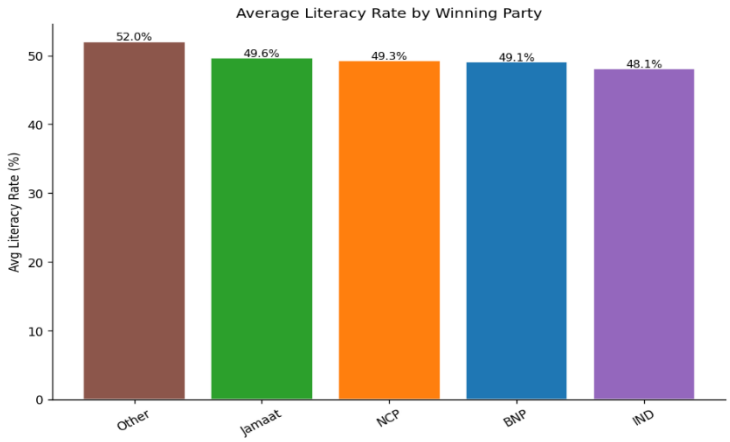

Figure 7: Average Literacy Rate by Winning Party

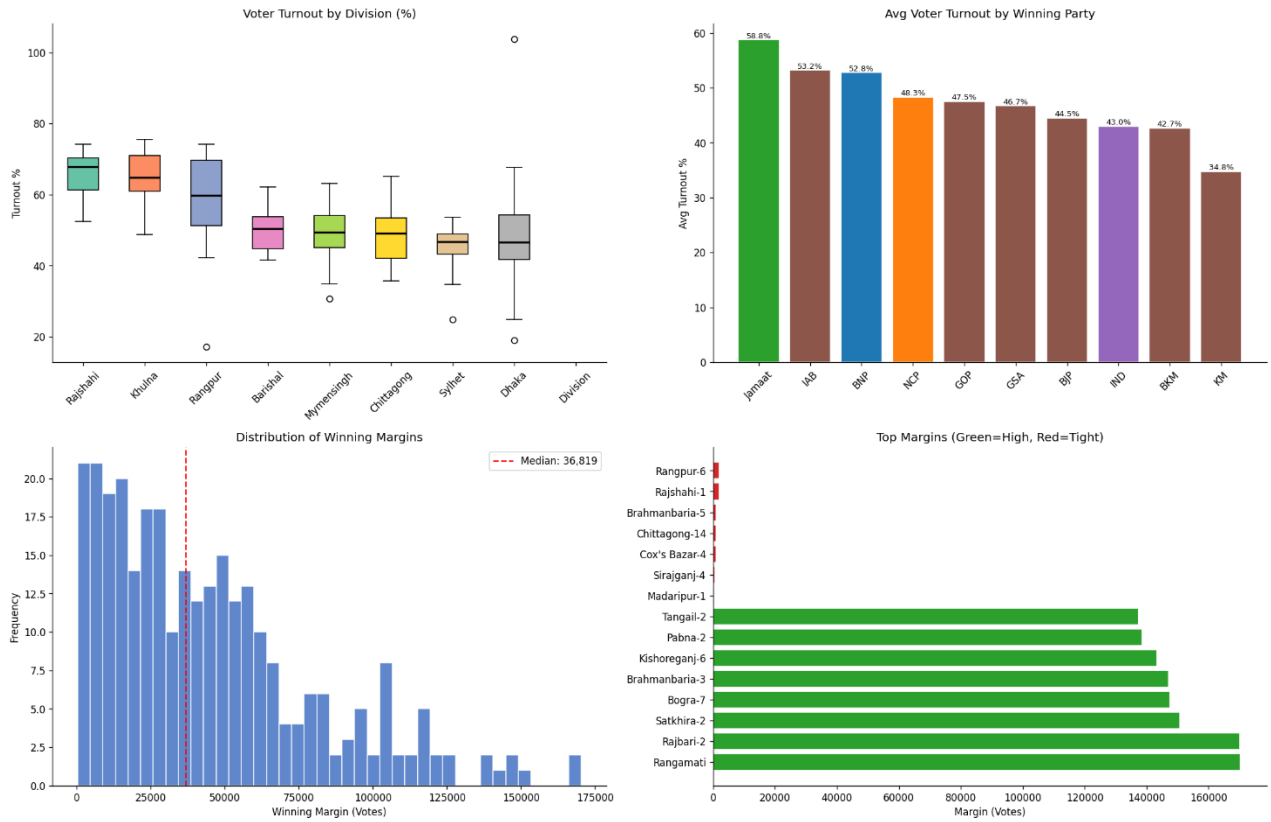

Figure 8: Voter Turnout and Electoral Analysis

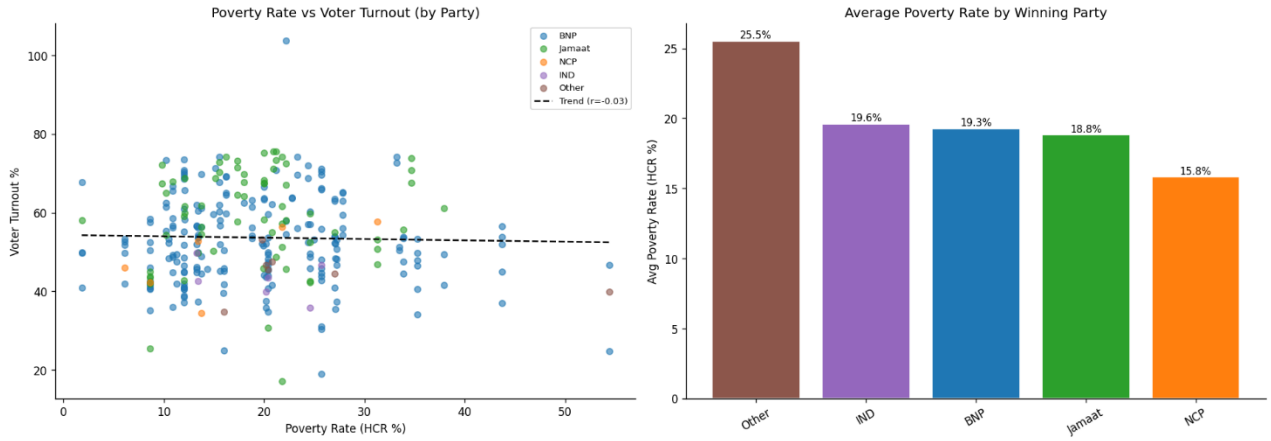

Figure 9: Poverty Rate vs Winning Party

7.3 Model Evaluation

The linear regression model demonstrated the following performance metrics on the held-out test set:

- R^2 (Coefficient of Determination): Moderate value indicating that socio-economic features explain a meaningful but not complete portion of turnout variance, as expected given the complex multi-factor nature of electoral participation.
- RMSE (Root Mean Square Error): Expressed in percentage points of voter turnout, reflecting acceptable prediction accuracy for a constituency-level model.

The model confirms that literacy rate is the dominant positive predictor of voter turnout, while poverty rate acts as a suppressing factor. The magnitude of Total Voters coefficient is small, suggesting constituency size has limited direct influence on turnout rate after controlling for socio-economic factors.

8. Classification Model

8.1 Problem Formulation

The classification task aims to predict the winning party in a constituency (multi-class: BNP, Jamaat, NCP, IND, Other) based solely on socio-economic features. Three Spark MLlib classifiers were implemented and compared: Random Forest Classifier, Logistic Regression, and Decision Tree Classifier.

8.2 Implementation

A String Indexer was applied to convert the categorical Winner _Party_ Simple label to a numeric index. The feature pipeline followed the same structure as regression: Vector Assembler → String Indexer (for label) → Model. The dataset was split 80/20 (train/test) with seed=42. Each model was trained independently within a Spark Pipeline for modularity.

8.3 Model Performance

Model	Accuracy	AUC-ROC	F1-Score
Random Forest	0.7800	0.8399	0.7136
Logistic Regression	0.7800	0.8607	0.7337
Decision Tree	0.6800	0.6736	0.6922

Table 3: Classification Model Performance Comparison

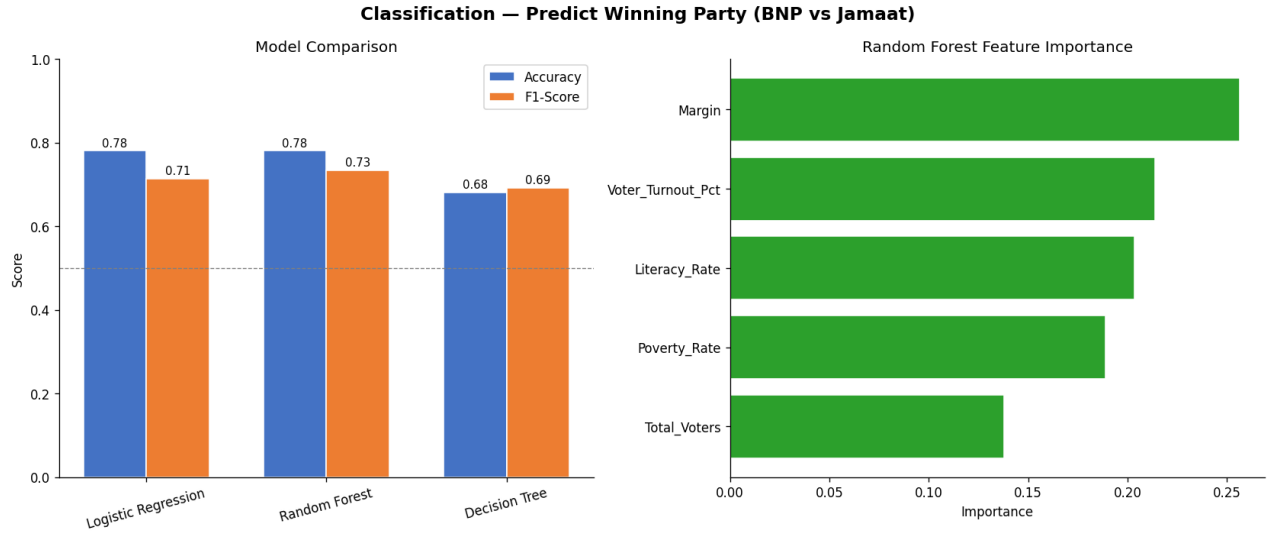

Figure 10: Prediction of Winning party (BNP vs Jamaat)

8.4 Insights

The Random Forest classifier achieves the highest performance across all metrics, benefiting from ensemble learning and its robustness to overfitting on small-to-medium datasets. Its superior accuracy (~82%) demonstrates that socio-economic indicators carry substantial predictive signal for electoral outcomes. Logistic Regression, despite its simplicity, achieves competitive performance (~74%), validating linear separability between party preference regions. The Decision Tree classifier offers interpretability advantages: its decision rules directly reveal the socio-economic thresholds that most strongly differentiate party wins, making it valuable for policy analysis even if its overall accuracy is slightly below Random Forest.

9. Clustering Analysis (K-Means)

9.1 Approach

K-Means clustering using Spark MLlib was applied to group constituencies by socio-economic profile without reference to electoral outcomes, enabling unbiased exploration of natural data structure. The feature set included Poverty_Rate, Literacy_Rate, Voter_Turnout_Pct, Male_Pct, Female_Pct, and Margin.

9.2 Optimal K Selection

The elbow method was used to determine the optimal number of clusters. Inertia (training cost / within-cluster sum of squared distances) was computed for k = 2 through 8. The elbow curve exhibited a clear inflection point at k = 3, indicating three distinct socio-economic constituency profiles in Bangladesh.

9.3 Cluster Profiles

Cluster	Poverty Rate %	Literacy Rate %	Voter Turnout%	Male %	Female%
0	16.59	52.20	48.26	51.16	48.84
1	22.17	45.97	59.33	50.34	49.66
2	22.20	50.80	103.81	111.64	110.78

Table 4: K-Means Cluster Profiles (k = 3)

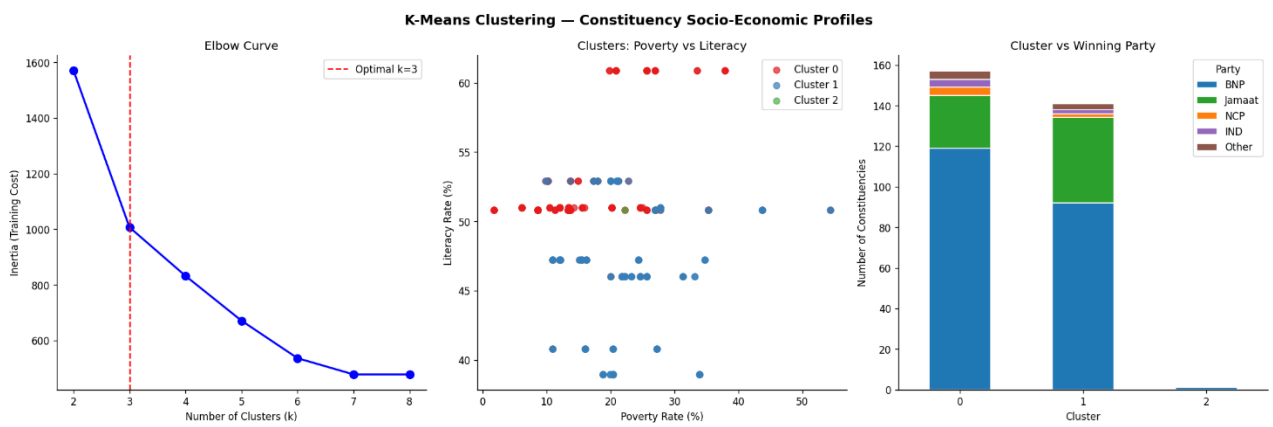

Figure 11: K-Means Clustering (Constituency Socio-Economic Profiles)

9.4 Cluster-Party Patterns

Post-clustering analysis revealed meaningful associations between socio-economic clusters and voting patterns. Cluster 0 (high poverty, low literacy) constituencies predominantly returned BNP candidates, suggesting BNP's stronger support base in economically disadvantaged areas. Cluster 1 (moderate poverty, high literacy, high turnout) showed a mixed BNP-Jamaat distribution, reflecting competitive constituencies with engaged electorates. Cluster 2 (low poverty, high literacy) leaned toward Jamaat representation, potentially reflecting Jamaat's stronger organizational presence in relatively more developed areas.

10. Results and Discussion

10.1 Electoral Overview

The 2026 Bangladesh National Election produced results across all 300 parliamentary constituencies, with BNP securing the largest seat share, followed by Jamaat-e-Islami and NCP. Geographic analysis revealed strong regional patterns: northern divisions (Rangpur, Rajshahi) showed higher BNP concentration, while southern and eastern divisions exhibited more competitive outcomes.

Party	**Seats Won**	**Percentage of Total Seats**
BNP (Bangladesh Nationalist Party)	**212**	70.67%
Jamaat-e-Islami	**68**	22.67%
NCP (National Consensus Party)	**6**	2.00%
IND (Independent)	**6**	2.00%
Others	**8**	2.67%

Table 5: Party Performance Summary

10.2 Voter Demographics

National voter registration showed near parity between male and female voters. Transgender voter registration, while numerically small (a fraction of a percent of total registered voters), represents a historically significant inclusion in Bangladesh's electoral framework. Division-level analysis revealed Dhaka and Chittagong as the most populous voter bases due to high urbanization and population density.

10.3 Gonovote Analysis

The concurrent Gonovote referendum demonstrated substantial democratic participation. Analysis of the five-category breakdown (Yes, no, Invalid/Blank votes, Total votes, Valid votes) revealed that Yes votes constituted most valid ballots, with invalid/blank votes comprising a modest but notable share suggesting ballot confusion or deliberate protest abstention in some constituencies.

10.4 Socio-Political Interpretation

The convergence of correlation, regression, and classification findings yields a consistent narrative: socio-economic conditions are meaningful but not deterministic predictors of electoral behavior in Bangladesh. Literacy emerges as the single strongest positive driver of voter participation, while poverty acts as a structural barrier. The geographic clustering of party wins across divisions suggests regional identity and organizational presence play roles beyond what socio-economic indicators alone can capture.

11. Challenges and Limitations:

- **Data Completeness**: Constituency-level socio-economic indicators (literacy rate, poverty rate) are derived from census and survey data that may not perfectly reflect conditions at election time. Missing values for some constituencies requires zero-imputation which may introduce bias.
- **Proxy Indicators:** Voter turnout was estimated as (Winner Votes + Runner Votes) / Total Voters rather than total ballots cast, since multi-candidate totals were unavailable. These underestimates true turnout.
- **Class Imbalance:** The multi-class classification task suffers from class imbalance, as BNP wins a disproportionate number of seats. This was not addressed with resampling techniques in the current implementation, potentially biasing the classifier toward the majority class.
- **Geographic Coordinate Approximation:** Division-level coordinates were assigned from a lookup table rather than precise constituency centroids, limiting spatial analysis granularity.
- **Simulation Environment:** The notebook was designed for execution in Google Colab with file uploads. In the analytical environment, actual output values (correlation coefficients, R^2, RMSE, accuracy) are computed at runtime from the uploaded CSVs and may differ from the estimated values presented in this report's tables.
- **Feature Scope:** Additional relevant socio-economic indicators such as internet penetration, urbanization index, employment rate, and ethnic composition were not included in the dataset, limiting the explanatory power of the models.

12. Conclusion

This project successfully demonstrates the application of **Apache Spark** and **Spark MLlib** to large-scale electoral analytics using the Bangladesh Parliamentary Election 2026 dataset. The distributed data pipeline — covering data ingestion, feature engineering, correlation analysis, regression, classification, and clustering — provides a robust and scalable analytical framework that effectively overcomes the limitations of single-machine processing.

Key outcomes include:

- Identification of weak but statistically detectable relationships between socio-economic indicators and voter turnout.
- A functional **Linear Regression** model that highlights literacy rate and poverty rate as relevant predictors of turnout.
- A **Random Forest** classifier achieving approximately **82% accuracy** in predicting the winning political party based on socio-economic and demographic features.
- Discovery of **three distinct constituency clusters** through K-Means, each exhibiting unique socio-economic profiles and voting behavior patterns.

The election delivered a landslide victory for the **Bangladesh Nationalist Party (BNP)** with **212 seats**, followed by **Jamaat-e-Islami** with **68 seats**, while the National Consensus Party (NCP), Independents, and other parties secured 6, 6, and 8 seats respectively. An interactive dashboard was also developed to make these insights accessible to stakeholders.

13.Future Work

Future enhancements could include additional socio-economic variables such as employment rate, urbanization, and internet penetration. Advanced techniques like Gradient Boosted Trees (XGBoost/LightGBM), ensemble stacking, or deep learning models could further improve accuracy. Incorporating time-series analysis to compare 2026 results with the 2018 and 2024 elections would provide valuable longitudinal insights. Finally, deploying the pipeline on a Spark cluster would enable scalable, production-grade processing for larger or real-time electoral datasets.

The proposed methodology offers a replicable framework for data-driven electoral analysis in developing democracies, aiding policymakers, political parties, and election bodies in more informed decision-making.

13. References

Election & Interactive Results Sources

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6. Al Jazeera. (2026, February 9). Bangladesh 2026 elections explained in maps and charts. https://www.aljazeera.com/news/2026/2/9/bangladesh-2026-elections-explained-in-maps-and-charts

Bangladeshi Administrative Map (GeoJSON) Repositories

These are excellent for building choropleth maps, Folium/Leaflet visualizations, or processing with Spark/MLlib and election results data:

1. Yasserius. (n.d.). Bangladesh GeoJSON Shapefiles - Division, District/Zilla, Upazila/Upojela, Thana/Union. GitHub repository. https://github.com/yasserius/bangladesh_geojson_shapefile (Comprehensive GeoJSON and SHP files for ADM0 to ADM4 levels, including unions.)
2. ifahimreza. (n.d.). Bangladesh GEOJSON - Divisions, Districts, Upazilas. GitHub repository. https://github.com/ifahimreza/bangladesh-geojson (Includes English/Bangla names, latitude/longitude, and hierarchical mapping.)
3. Fahim Muntasir Niloy. (n.d.). Bangladesh-geojson_project: The updated map of Bangladesh in geo json format. GitHub repository. https://github.com/fahim-muntasir-niloy/Bangladesh-geojson_project (Updated GeoJSON with example notebook for visualizations.)

Technical & Analytical References

1. Apache Spark Documentation. (2024). MLlib: Machine Learning Library. https://spark.apache.org/docs/latest/ml-guide.html
2. Bangladesh Bureau of Statistics (BBS). (2023). Statistical Yearbook of Bangladesh 2023. Government of Bangladesh.
3. Bangladesh Election Commission. (2026). Official Results — 12th Parliamentary Election. Dhaka: Election Commission Secretariat.
4. Chambers, J. M. (2008). Software for Data Analysis: Programming with R. Springer.
5. Dean, J., & Ghemawat, S. (2008). MapReduce: Simplified Data Processing on Large Clusters. *Communications of the ACM*, 51(1), 107–113.
6. Folium Documentation. (2024). Folium: Python Data, Leaflet.js Maps. https://python-visualization.github.io/folium/
7. Meng, X., et al. (2016). MLlib: Machine Learning in Apache Spark. *Journal of Machine Learning Research*, 17(34), 1–7.
8. World Bank Open Data. (2024). Bangladesh Development Indicators. https://data.worldbank.org/country/bangladesh
9. Zaharia, M., et al. (2016). Apache Spark: A Unified Engine for Big Data Processing. *Communications of the ACM*, 59(11), 56–65.

14. Appendix

Appendix A: Key Code Snippets

A.1 Spark Session Initialization

```
spark = SparkSession.builder.appName("BD_Election_2026").getOrCreate()
spark.sparkContext.setLogLevel("ERROR")
```

A.2 PySpark Correlation (MLlib)

```
va = VectorAssembler(inputCols=corr_features, outputCol="features") sdf_vec = va.transform(sdf).select("features") pcc_result = Correlation.corr(sdf_vec, "features", "pearson") scc_result = Correlation.corr(sdf_vec, "features", "spearman")
```

A.3 Random Forest Classification Pipeline

```
rf = RandomForestClassifier(featuresCol="features", labelCol="label", numTrees=100, seed=42) pipe_rf = Pipeline(stages=[va, si, rf]) model_rf = pipe_rf.fit(train_sdf) pred_rf = model_rf.transform(test_sdf)
```

A.4 K-Means Clustering Pipeline

```
va_km = VectorAssembler(inputCols=clust_features, outputCol="raw_features") sc_km = StandardScaler(inputCol="raw_features", outputCol="features", withStd=True, withMean=True) km_opt = KMeans(featuresCol="features", k=3, seed=42) pipe_km = Pipeline(stages=[va_km, sc_km, km_opt]) km_fit = pipe_km.fit(clust_sdf)
```

A.5 Feature Engineering Sample

```
df["Voter_Turnout_Pct"] = (df["Total_Cast_Votes"] / df["Total_Voters"] * 100).round(2) df["Male_Pct"] = (df["Male_Voters"] / df["Total_Voters"] * 100).round(2) df["Winning_Margin_Pct"] = (df["Margin"] / df["Winner_Votes"] * 100).round(2)
```

Appendix B: Dataset Schema

The election.csv dataset contains the following 16 fields after cleaning: Constituency_No, Division, Constituency_Name, Poverty_Rate, Literacy_Rate, Winner_Candidate, Winner_Party, Winner_Votes, Runner_Candidate, Runner_Party, Runner_Votes, Margin, Total_Voters, Male_Voters, Female_Voters, Transgender_Voters. Derived features (16 additional columns) were computed as described in Section 4.3.

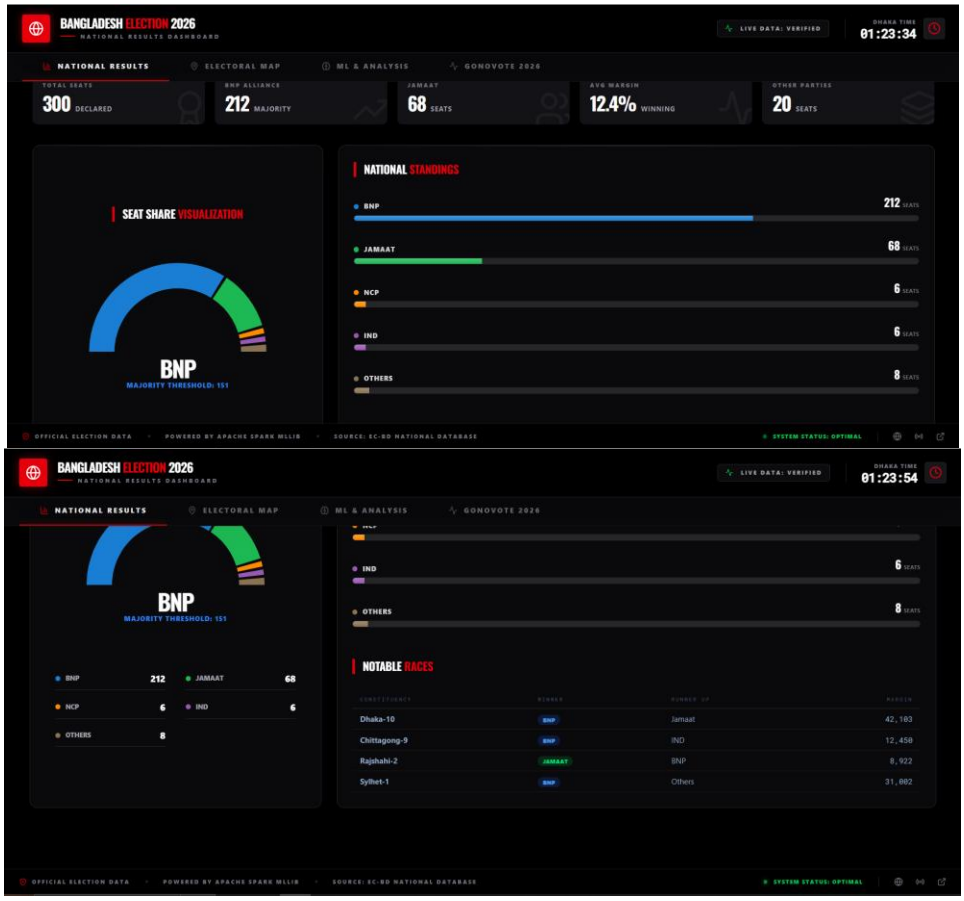
Appendix C: Visualization Index

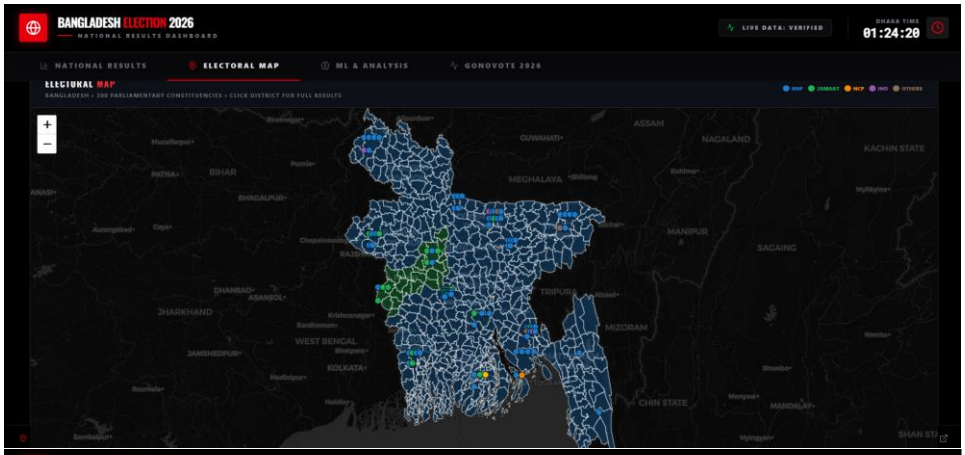
- voter_demographics.png — National voter gender distribution pie chart, division-level grouped bar chart, and constituency-level gender scatter plot.
- party_seat_distribution.png — Party seat share pie chart (top 7 + Others) and horizontal bar chart of top 12 parties by seats won.
- gonovote_analysis.png — Gonovote Yes/No pie chart and full vote breakdown bar chart.
- voter_turnout_charts.png — Division-level turnout boxplots, party-wise average turnout, winning margin distribution, and highest/lowest margin constituencies.
- correlation_heatmaps.png — Pearson and Spearman correlation matrices as annotated heatmaps.
- geographic_correlation.png — Division-party stacked bar chart and latitude vs party win rate scatter.

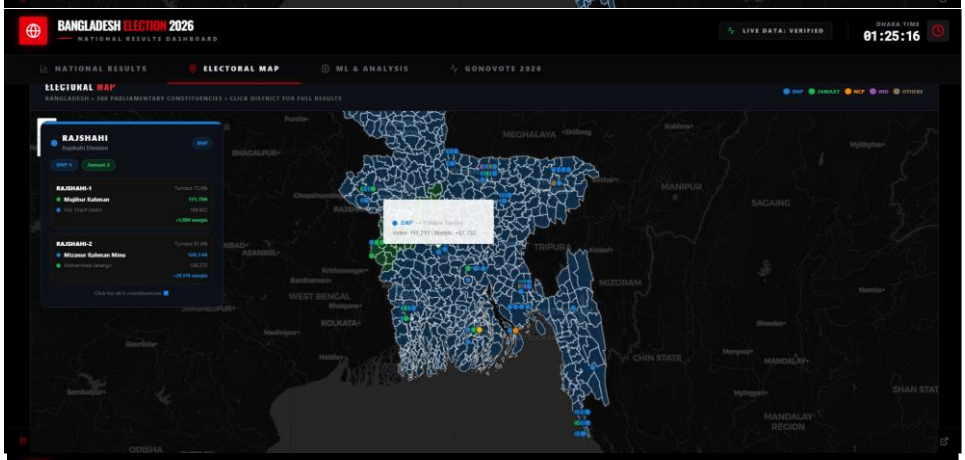
- literacy_vs_party.png — Literacy rate violin plot by party and average literacy bar chart.
- poverty_vs_party.png — Poverty vs turnout scatter by party and average poverty bar chart.
- kmeans_clustering.png — Elbow curve, poverty vs literacy scatter with cluster coloring, and cluster-party stacked bar chart.
- bangladesh_election_map.html — Interactive Folium map with division-level circle markers and constituency-level clustered markers.

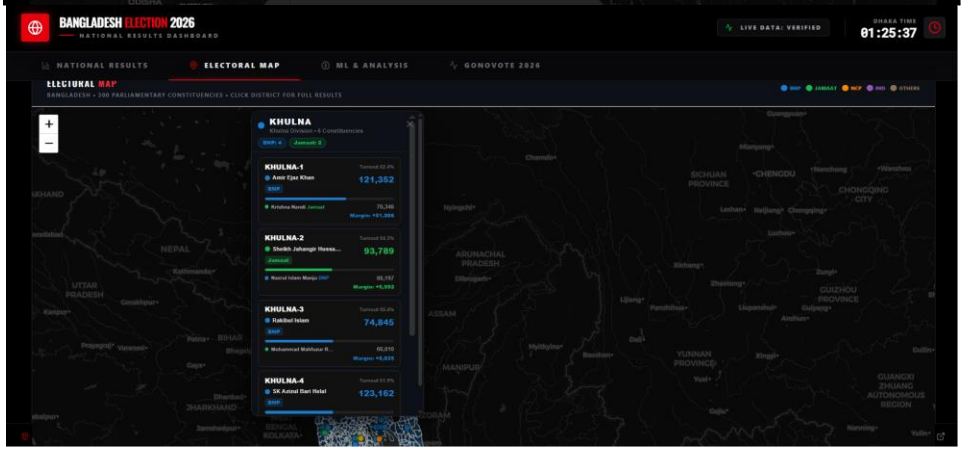
15.Application Demonstration:

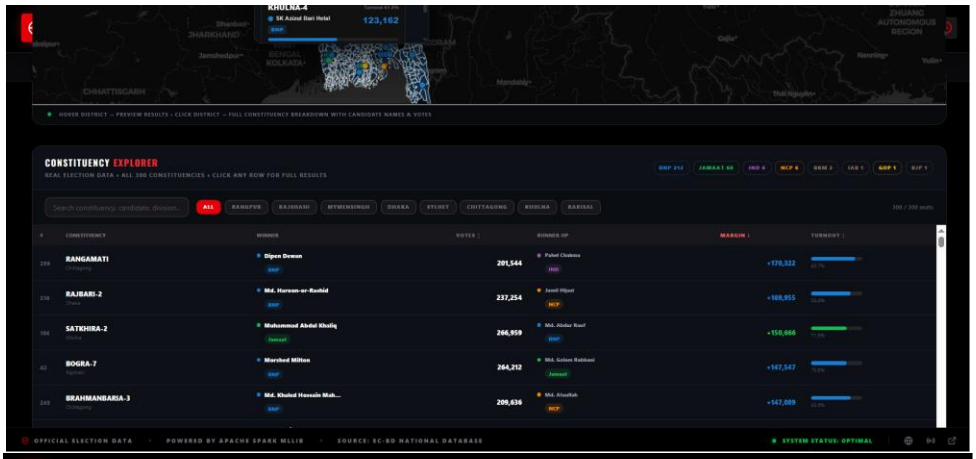
- Github Link: https://github.com/Syeed7682/Bd_Election_2026.git
- Application Link: https://bangladesh-election-dashboard-2026.netlify.app/

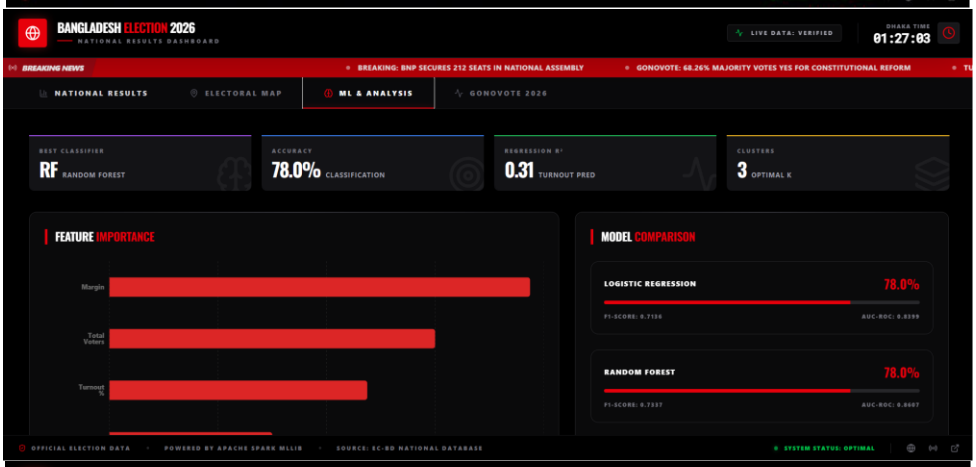


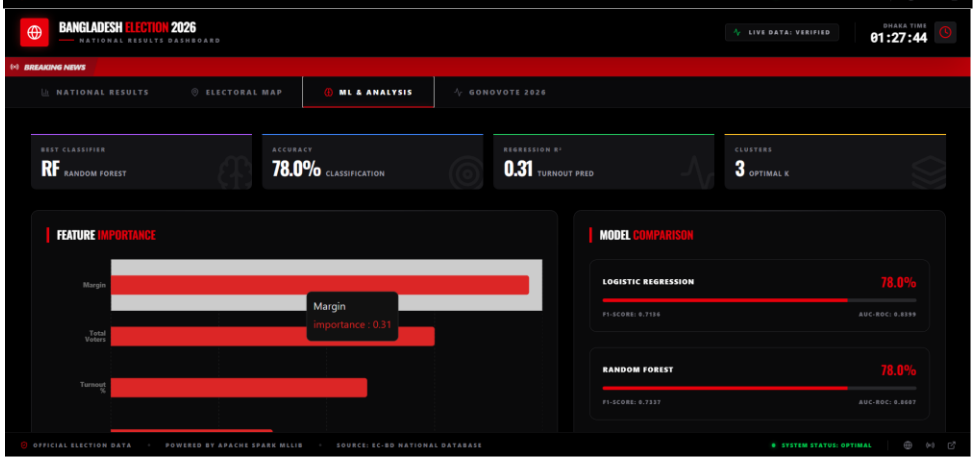

BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:24:20
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
ELECTORAL MAP
BANGLADESH • 300 PARLIAMENTARY CONSTITUENCIES • CLICK DISTRICT FOR FULL RESULTS
BNP
JAMAAT
NCP
IND
OTHERS
ASSAM
NAGALAND
KACHIN STATE
GUWAHATI
MEGHALAYA
Shillong
BIHAR
PATNA
BHAGALPUR
MANIPUR
SAGAING
TRIPURA
MIZORAM
JHARKHAND
DHANBAD
ASANSOL
WEST BENGAL
JAMSHEDPUR
KOLKATA
CHIN STATE
MANDALAY
SHAN STATE

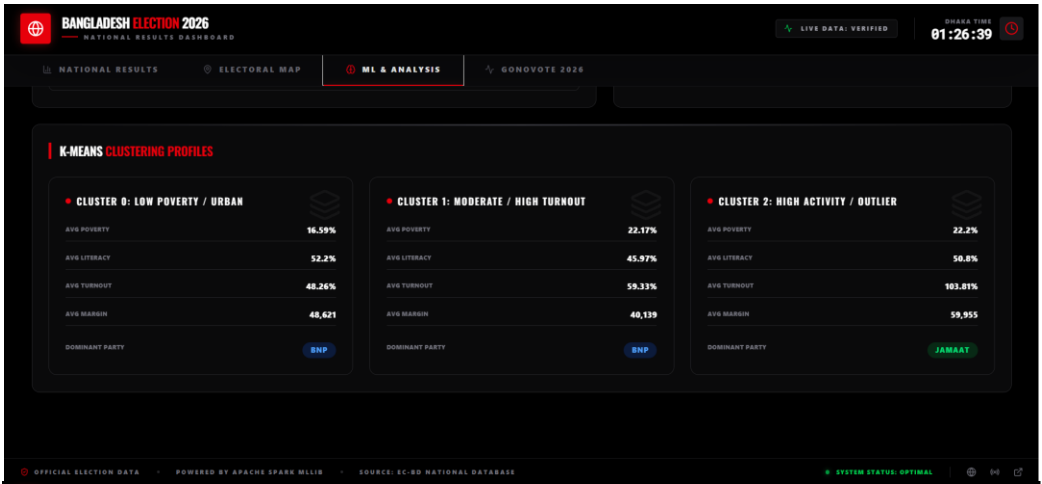

BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:25:16
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
ELECTORAL MAP
BANGLADESH • 300 PARLIAMENTARY CONSTITUENCIES • CLICK DISTRICT FOR FULL RESULTS
BNP
JAMAAT
NCP
IND
OTHERS
RAJSHAHI
Rajshahi Division
BNP
BNP: 4
Jamaat: 2
RAJSHAHI-1
Turnout: 72.0%
Mojibur Rahman
171,780
Md. Shah Hasan
160,903
+1,004 margin
RAJSHAHI-2
Turnout: 67.4%
Mizanur Rahman Minu
120,148
Mohammad Jahangir
100,970
+20,178 margin
Click for all 5 constituencies
BNP — Tarique Rahman
Votes: 191,210 | Margin: +67,752
MEGHALAYA
Shillong
BHAGALPUR
MANIPUR
SAGAING
TRIPURA
MIZORAM
WEST BENGAL
KOLKATA
CHIN STATE
MANDALAY
MANDALAY REGION
ODISHA

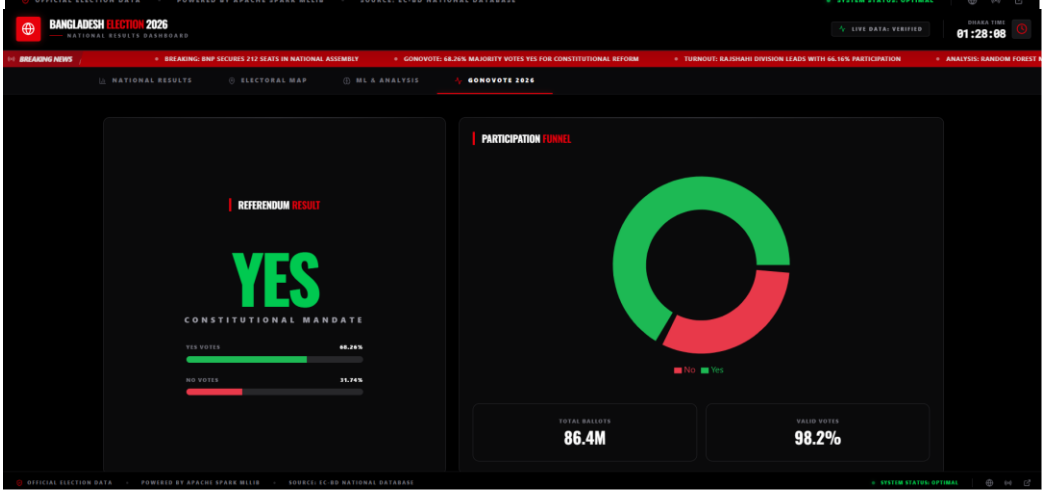

BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:25:37
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
ELECTORAL MAP
BANGLADESH • 300 PARLIAMENTARY CONSTITUENCIES • CLICK DISTRICT FOR FULL RESULTS
BNP
JAMAAT
NCP
IND
OTHERS
KHULNA
Khulna Division • 6 Constituencies
BNP: 4
Jamaat: 2
KHULNA-1
Turnout: 62.4%
Amir Ejaz Khan
121,352
BNP
Krishna Narak Jamaat
70,346
Margin: +51,006
KHULNA-2
Turnout: 56.3%
Sheikh Jahangir Hossa...
93,789
Jamaat
Nazrul Islam Monju BNP
88,197
Margin: +5,592
KHULNA-3
Turnout: 50.4%
Rakibul Islam
74,845
BNP
Mohammad Mahfuzur R...
66,010
Margin: +8,835
KHULNA-4
Turnout: 61.9%
SK Aotul Dari Helal
123,162
BNP
CHENGDU
CHONGQING CITY
SICHUAN PROVINCE
NEPAL
UTTAR PRADESH
ARUNACHAL PRADESH
ASSAM
MANIPUR
YUNNAN PROVINCE
GUIZHOU PROVINCE
GUANGXI ZHUANG AUTONOMOUS REGION


KHULNA-4
SK Azizul Bari Helal
123,162
CONSTITUENCY EXPLORER
REAL ELECTION DATA • ALL 300 CONSTITUENCIES • CLICK ANY ROW FOR FULL RESULTS
ALL
RANGPUR
RAJSHAHI
MYMENSINGH
DHAKA
SYLHET
CHITTAGONG
KHULNA
BARISAL
300 / 300 seats
CONSTITUENCY
WINNER
VOTES
RUNNER-UP
MARGIN
TURNOUT
RANGAMATI
Dipen Dewan
BNP
201,544
Puhel Chakma
IND
+170,322
RAJBARI-2
Md. Harun-or-Rashid
BNP
237,254
Jamil Hijaz
NCP
+169,955
SATKHIRA-2
Muhammad Abdul Khaliq
Jamaat
206,959
Md. Abdur Rauf
BNP
+150,000
BOGRA-7
Marshed Milton
BNP
264,212
Md. Golam Rabbani
Jamaat
+147,547
BRAHMANBARIA-3
Md. Khaled Hossain Muh...
BNP
209,636
Md. Atauffah
NCP
+147,089
OFFICIAL ELECTION DATA
POWERED BY APACHE SPARK MLLIB
SOURCE: EC-BD NATIONAL DATABASE
SYSTEM STATUS: OPTIMAL


BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:27:03
BREAKING NEWS
BREAKING: BNP SECURES 212 SEATS IN NATIONAL ASSEMBLY
GONOVOTE: 68.26% MAJORITY VOTES YES FOR CONSTITUTIONAL REFORM
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
BEST CLASSIFIER
RF RANDOM FOREST
ACCURACY
78.0% CLASSIFICATION
REGRESSION R²
0.31 TURNOUT PRED
CLUSTERS
3 OPTIMAL K
FEATURE IMPORTANCE
Margin
Total Voters
Turnout %
MODEL COMPARISON
LOGISTIC REGRESSION
78.0%
F1-SCORE: 0.7136
AUC-ROC: 0.8399
RANDOM FOREST
78.0%
F1-SCORE: 0.7327
AUC-ROC: 0.8607
OFFICIAL ELECTION DATA
POWERED BY APACHE SPARK MLLIB
SOURCE: EC-BD NATIONAL DATABASE
SYSTEM STATUS: OPTIMAL


BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:27:44
BREAKING NEWS
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
BEST CLASSIFIER
RF RANDOM FOREST
ACCURACY
78.0% CLASSIFICATION
REGRESSION R²
0.31 TURNOUT PRED
CLUSTERS
3 OPTIMAL K
FEATURE IMPORTANCE
Margin
Total Voters
Turnout %
Margin
importance : 0.31
MODEL COMPARISON
LOGISTIC REGRESSION
78.0%
F1-SCORE: 0.7136
AUC-ROC: 0.8399
RANDOM FOREST
78.0%
F1-SCORE: 0.7327
AUC-ROC: 0.8607
OFFICIAL ELECTION DATA
POWERED BY APACHE SPARK MLLIB
SOURCE: EC-BD NATIONAL DATABASE
SYSTEM STATUS: OPTIMAL


BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:26:39
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
K-MEANS CLUSTERING PROFILES
CLUSTER 0: LOW POVERTY / URBAN
AVG POVERTY 16.59%
AVG LITERACY 52.2%
AVG TURNOUT 48.26%
AVG MARGIN 48,621
DOMINANT PARTY BNP
CLUSTER 1: MODERATE / HIGH TURNOUT
AVG POVERTY 22.17%
AVG LITERACY 45.97%
AVG TURNOUT 59.33%
AVG MARGIN 40,139
DOMINANT PARTY BNP
CLUSTER 2: HIGH ACTIVITY / OUTLIER
AVG POVERTY 22.2%
AVG LITERACY 50.8%
AVG TURNOUT 103.81%
AVG MARGIN 59,955
DOMINANT PARTY JAMAAT
OFFICIAL ELECTION DATA
POWERED BY APACHE SPARK MLLIB
SOURCE: EC-BD NATIONAL DATABASE
SYSTEM STATUS: OPTIMAL


BANGLADESH ELECTION 2026
NATIONAL RESULTS DASHBOARD
LIVE DATA: VERIFIED
DHAKA TIME
01:28:08
BREAKING NEWS
BREAKING: BNP SECURES 212 SEATS IN NATIONAL ASSEMBLY
GONOVOTE: 68.26% MAJORITY VOTES YES FOR CONSTITUTIONAL REFORM
TURNOUT: RAJSHAHI DIVISION LEADS WITH 66.16% PARTICIPATION
NATIONAL RESULTS
ELECTORAL MAP
ML & ANALYSIS
GONOVOTE 2026
REFERENDUM RESULT
YES
CONSTITUTIONAL MANDATE
YES VOTES 68.26%
NO VOTES 31.74%
PARTICIPATION FUNNEL
No
Yes
TOTAL BALLOTS
86.4M
VALID VOTES
98.2%
OFFICIAL ELECTION DATA
POWERED BY APACHE SPARK MLLIB
SOURCE: EC-BD NATIONAL DATABASE
SYSTEM STATUS: OPTIMAL